

Language vs. Law: A Neural Alignment Adapter for Taglish Legal Query Retrieval in Philippine Statutes

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ABSTRACT

The use of Taglish in Philippine legal research creates a register mismatch with formal English statutes that hinders information retrieval. This study introduces the Philippine Legal Retrieval Benchmark (PLRB)—comprising 2,400 source documents, 28,970 passages, and 180 researcher-constructed queries (60 each in English, Tagalog, and Taglish)—and evaluates three frozen encoder backbones, mSBERT, BGE-M3, and Legal-BERT, under both a zero-shot baseline retrieval setting and a query-side adapter-aligned setting. Over the 29-query evaluated subset for which relevance judgments exist, BGE-M3 attains the strongest zero-shot performance with an overall MRR of 0.8563 and a P@5 of 0.5310, including a perfect MRR of 1.0000 on the single code-switched (Taglish) query in the subset. The attached evaluation results do not show a global improvement for the adapter-aligned setting: BGE-M3's overall MRR falls to 0.5607, mSBERT's falls to 0.3852, and Legal-BERT's falls to 0.0201. The one positive signal is mSBERT's Taglish MRR, which rises from 0.0000 to 0.2500 under alignment, though this is based on a single query and should be read as a case-level observation rather than a stable trend. The final reported retrieval system is therefore the zero-shot BGE-M3 baseline; the adapter-aligned setting is reported for completeness and analysis but is not recommended for deployment under current data conditions. These findings indicate that the adapter overfits the small training split, and that zero-shot multilingual encoders—particularly BGE-M3—remain the most viable retrieval option for Philippine legal queries given present resource constraints.

CCS Concepts

- Information systems → Information retrieval;
Computing methodologies → Natural language processing.

Keywords

- Neural alignment adapter, contrastive learning, triplet margin loss, code-switching (Taglish), Philippine legal NLP, semantic search, dense retrieval, FAISS.

1. INTRODUCTION

The Philippine legal system codifies its statutes and court decisions in formal English, yet citizens, students, and non-specialist practitioners typically search for legal information

in Taglish—a code-switched register that blends Tagalog morphology with English syntax. When a user asks, “Ano ang mga parusa sa estafa sa ilalim ng Revised Penal Code?”, they express a precise legal concern in a natural, familiar register, but the indexed source material is written in formal English statutory prose. This produces a register mismatch that is distinct from the general difficulty of multilingual retrieval: even an encoder that handles English, Tagalog, and code-switched text individually may still place a Taglish query and its relevant formal-English passage far apart in embedding space, because the two differ in register and formality rather than only in language identity.

The prevailing approach—using off-the-shelf multilingual encoders such as mSBERT and BGE-M3 in zero-shot mode—assumes that broad multilingual pretraining is sufficient to bridge this register gap. That assumption has rarely been tested on Philippine legal corpora. Legal-domain models such as Legal-BERT excel at formal English legal language but were pretrained exclusively on English text and have no exposure to Tagalog or Taglish. Neither class of model has been evaluated on the specific problem of code-switched Taglish queries retrieving against formal English Philippine statutory and jurisprudential passages. This gap is the central problem this study addresses.

Research Question: Does a parameter-efficient, query-side neural alignment adapter improve retrieval of formal English Philippine legal passages from English, Tagalog, and Taglish queries, relative to zero-shot multilingual and legal-domain baselines, as measured by MRR, Precision@5, Precision@10, and Recall@10 on the Philippine Legal Retrieval Benchmark (PLRB)?

This study makes three distinct contributions, which we separate explicitly to avoid conflating them. First, a new benchmark: the Philippine Legal Retrieval Benchmark (PLRB), the first publicly documented retrieval benchmark covering English, Tagalog, and Taglish legal queries against a Philippine statutory and jurisprudential passage index. Second, a new adapter: a lightweight, query-side-only neural alignment adapter—a single linear projection followed by LayerNorm and ReLU—trained with triplet margin loss on top of frozen encoder embeddings, designed to avoid re-encoding the 28,970-passages FAISS index. Third, a new evaluation pipeline: a modular FAISS-based retrieval and evaluation pipeline that reports MRR, P@5, P@10, and Recall@10 broken down by query language and semantic type, applied consistently to both the zero-shot and adapter-aligned settings. The retrieval system evaluated in this study is a

single-stage dense retrieval pipeline; no BM25 or sparse-reranking stage was implemented or evaluated.

The adapter-based approach is suited to three practical constraints common to local Philippine academic institutions: full fine-tuning of large multilingual encoders is computationally prohibitive on typical undergraduate hardware, labeled Philippine legal retrieval data is scarce, and any deployed system must handle English, Tagalog, and Taglish queries within a single retrieval index without maintaining separate per-language pipelines.

Contributions of this study:

- PLRB, the Philippine Legal Retrieval Benchmark: 2,400 source documents, 28,970 retrieval passages, and 180 researcher-constructed queries (60 per language, stratified by semantic type), with LLM-assisted relevance labels reviewed by the full research team.
- A neural alignment adapter consisting of a linear projection, LayerNorm, and ReLU activation, trained with triplet margin loss on Philippine Bar Examination-informed query-passage pairs, applied only to query embeddings.
- A query-side-only adaptation strategy that avoids the overhead of re-encoding all 28,970 passage embeddings and reconstructing the FAISS IndexFlatIP structure when the adapter is updated.
- A modular FAISS-based evaluation pipeline with per-language and per-semantic-type breakdowns (MRR, P@5, P@10, Recall@10) applied uniformly to the baseline and aligned settings.
- Empirical evidence on adapter behavior, including the finding that the adapter reduces overall MRR, P@5, P@10, and Recall@10 for all three encoders relative to their zero-shot baselines, while lifting mSBERT's Taglish MRR from 0.0000 to 0.2500 on a single evaluated Taglish query.

PLRB was developed under undergraduate thesis constraints: queries were authored by the research team, and relevance labels were produced through LLM-assisted annotation with full team review rather than independent expert adjudication; Section 7 discusses these limitations in detail. The remainder of the paper is organized as follows: Section 2 reviews related literature across legal NLP, dense retrieval, code-switching, and Philippine NLP; Section 3 presents the methodology, including the problem definition, baseline models, adapter architecture, PLRB construction, and evaluation protocol; Section 4 describes the implementation, tools, and development process; Section 5 presents results and discussion; Section 6 concludes and outlines future work; and Section 7 details the limitations of the study.

2. REVIEW OF RELATED LITERATURE

This section synthesizes prior work across four thematic clusters that jointly motivate the present study: legal NLP and information retrieval, dense retrieval and contrastive learning, code-switching research, and Philippine NLP. Each cluster identifies the state of the art and the specific gap that this study targets.

2.1 Legal NLP and Information Retrieval

Legal information retrieval has matured from classical sparse, term-matching methods toward neural architectures, but the literature converges on a single theme: domain specificity matters, and general-purpose models transfer poorly without adaptation. Chalkidis et al. (2022) show that transformer-based models achieve strong performance on EU legislative classification and summarization, establishing that transformers are viable for structured legal text once trained on in-domain data. Thakur et al. (2021) complicate this picture with the BEIR benchmark, demonstrating that models trained on general-domain corpora often underperform on specialized retrieval tasks—a warning that is directly relevant to applying off-the-shelf encoders to Philippine legal text without validation. Kim et al. (2024) report that transformer-based embeddings outperform BM25 on the COLIEE legal retrieval benchmark, particularly for case-law queries that require semantic generalization rather than exact lexical matching, reinforcing the case for dense retrieval over sparse methods in legal domains. Voorhees (2001) supplies the evaluation framework—MRR and Precision@K—that this study adopts, establishing these metrics as standard for ranked retrieval evaluation.

Read together, these studies establish that (a) transformers outperform sparse methods on legal retrieval when in-domain, (b) general-domain pretraining alone is an unreliable foundation for specialized retrieval, and (c) MRR and Precision@K are the appropriate metrics for this evaluation. None of these studies, however, examine retrieval over Philippine-language legal queries, and none address the specific register mismatch between informal Taglish queries and formal English Philippine statutes—the gap this study targets. This study implements a single-stage dense retrieval pipeline; it does not implement or evaluate a hybrid BM25-plus-reranker architecture, though Kim et al.'s (2024) finding that dense embeddings outperform BM25 on case-law queries motivates the dense-only design choice.

2.2 Dense Retrieval, Contrastive Learning, and Adaptation

Contrastively pretrained encoders (Wang et al., 2022) generalize strongly across general retrieval benchmarks, and multilingual encoders such as mSBERT and BGE-M3 extend this contrastive training to over 100 languages. BGE-M3's 8,192-token context window is particularly relevant for Philippine legal passages, which frequently exceed the 512-token limits of earlier encoders. Despite this breadth, Wang et al. (2022) and related contrastive-pretraining work do not specifically test register mismatch: a Taglish query and a formal statute addressing the same legal concept can still receive poorly aligned embeddings even after broad cross-lingual pretraining, because contrastive objectives during pretraining rarely include register-shifted positive pairs.

This is where low-resource and domain-adaptation literature becomes directly relevant. Ahmadi et al. (2026) show that domain-adapted multilingual embeddings consistently improve top-k retrieval accuracy for low-resource legal language variants, providing direct motivation for adapter-based domain adaptation in this study. Schroff et al. (2015), working in the image domain, establish triplet margin loss as an effective contrastive alignment objective; this study applies the same objective to query-passage pairs rather than image triplets. Adapter-based alignment—small trainable layers placed on top of frozen encoders—has been shown to approach full fine-tuning quality at a fraction of the computational cost, which is the property that makes it suitable

for the hardware constraints described in Section 4.1. Taken together, this cluster motivates the adapter design but also flags the central risk this study confirms empirically: Ahmadi et al.’s (2026) positive results were obtained with larger training sets than PLRB’s 180 queries, and the present results (Section 5) show that adapter alignment does not transfer when the training split is this small.

2.3 Code-Switching in NLP

Filipino-English code-switching (Taglish) follows well-documented syntactic patterns and serves as the natural informal register for inter-Filipino discourse (Martin, 2012). Cruz and Cheng (2019, 2021) demonstrate that models pretrained on Filipino text outperform multilingual baselines on Filipino classification and named entity recognition tasks, establishing that language-specific pretraining provides measurable gains over generic multilingual coverage for Filipino NLP. Subsequent work—Barrios et al. (2023) on Tagalog named entity recognition and Aquino and de Leon (2020) on Filipino NER corpus construction—expands corpus coverage and annotation resources for Filipino NLP but remains focused on classification and extraction tasks rather than retrieval, and none of these studies touch the legal domain.

The combined effect of these two observations—that Filipino-specific pretraining outperforms generic multilingual coverage on non-retrieval tasks, and that no prior work has tested multilingual encoders on legal-domain Taglish retrieval—leaves open the central empirical question this study investigates: whether general-purpose multilingual encoders such as BGE-M3 and mSBERT, in zero-shot mode, can handle the register and code-switching demands of Philippine legal Taglish queries, or whether some form of adaptation (such as the alignment adapter proposed here) is necessary.

2.4 Philippine Legal AI and Research Gaps

Institutional support for Philippine legal AI has grown in recent years, providing context for why a retrieval benchmark of this kind is timely. The Supreme Court’s Strategic Plan for Judicial Innovations 2022–2027 designates NLP-enhanced legal research as an institutional priority (Supreme Court of the Philippines, 2022b), and the Office of the Chief Justice has publicly announced plans for the judiciary’s e-library to incorporate AI-based search technology (Supreme Court of the Philippines, 2022a). At the agency level, DOST-ASTI’s iTANONG project develops Taglish natural-language interfaces for structured database querying (Department of Science and Technology – Advanced Science and Technology Institute, 2021), demonstrating institutional interest in Taglish natural-language interfaces, though iTANONG addresses structured database querying rather than retrieval over unstructured legal passages.

Despite this institutional momentum, three gaps remain unaddressed in the literature reviewed above. First, no prior published work has evaluated embedding-based dense retrieval specifically on Taglish legal queries against Philippine statutory and jurisprudential text. Second, no parameter-efficient alignment method has been evaluated for the Taglish-to-Philippine-legal-passage retrieval problem. Third, no publicly documented Philippine legal retrieval benchmark with per-language and per-semantic-type evaluation exists. This study addresses all three gaps by contributing the PLRB benchmark, evaluating a query-side alignment adapter against zero-shot baselines, and reporting stratified results—while also providing,

in Section 5, empirical evidence on the limits of lightweight adapter-based alignment under the low-resource conditions typical of undergraduate-thesis-scale data collection.

3. METHODOLOGY

3.1 Problem Definition

Let $Q = \{q_1, \dots, q_n\}$ be a set of queries drawn from English, Tagalog, or Taglish registers, and let $D = \{d_1, \dots, d_m\}$ be the passage index derived from Philippine legal documents, with $m = 28,970$. For each query q_i , the retrieval system returns a ranked list $R(q_i)$ of candidate passages from D . Performance is evaluated using Precision@K for $K \in \{5, 10\}$, Recall@10, and Mean Reciprocal Rank (MRR), each defined formally in Section 3.5. The core challenge is register mismatch: informal code-switched query embeddings are geometrically distant from formal English document embeddings in standard shared embedding spaces, even under broad multilingual pretraining, because register and formality differences are not the primary axis along which general-purpose multilingual encoders were trained to align representations.

3.2 Baseline Models

Three frozen encoder architectures spanning different pretraining regimes bracket the design space:

- **mSBERT (paraphrase-multilingual-mpnet-base-v2)** — a general-purpose multilingual encoder trained via knowledge distillation with coverage across 50+ languages and no legal-domain specialization. Serves as the weak multilingual baseline.
- **BGE-M3 (BAAI/bge-m3)** — the strongest zero-shot candidate, combining 8,192-token context capacity with broad cross-lingual transfer across 100+ languages. Serves as the upper-bound zero-shot baseline.
- **Legal-BERT (nlpueb/legal-bert-base-uncased)** — pretrained on 12 GB of English legal text, capturing legal terminology and formal register but entirely monolingual. Serves as the legal-domain specialist baseline.

All three encoders are held frozen throughout the study; no encoder weights are updated. Adapter training modifies only the lightweight alignment layer described below.

3.3 Neural Alignment Adapter

The primary technical contribution is a lightweight alignment adapter placed on top of each frozen encoder’s query representation. Given a query embedding $e_q \in \mathbb{R}^d$ produced by the frozen encoder, the adapter transforms it as:

$$\hat{e}_q = \text{ReLU}(\text{LayerNorm}(W \cdot e_q + b))$$

where $W \in \mathbb{R}^{(d \times d)}$ and $b \in \mathbb{R}^d$ are the only trainable parameters, d is the encoder’s native embedding dimension, and $\hat{e}_q$ is the aligned query embedding used for retrieval in the adapter-aligned setting. The adapter introduces no additional parameters to the passage encoding path: passage embeddings are generated directly by the frozen encoder, L2-normalized, and stored in the FAISS index without any adapter transformation. Only the query path is modified.

The adapter is trained using triplet margin loss over triplets (q, p^+, p^-) , where p^+ is a passage labeled relevant to query q and p^- is a passage not labeled relevant to q . The loss function is:

$$L = \max(0, d(\hat{e}_q, e_{p^+}) - d(\hat{e}_q, e_{p^-}) + \alpha)$$

where $d(\cdot, \cdot)$ denotes cosine distance, e_{p^+} and e_{p^-} are the frozen passage encoder's L2-normalized embeddings for p^+ and p^- respectively, and $\alpha = 0.3$ is the margin hyperparameter. The adapter is optimized using Adam with a learning rate of 2×10^{-4} for 50 epochs on the PLRB training split (Section 3.4). Negatives p^- were sampled uniformly at random from passages not labeled as relevant to query q , drawn from the same document-type pool as the query's positives; hard negative mining—selecting negatives that are embedding-similar to the query but unlabeled as relevant—was not applied in this study. As discussed in Section 5.2, this uniform sampling strategy is a candidate explanation for the adapter's limited discriminability against already-strong zero-shot encoders such as BGE-M3.

3.4 Philippine Legal Retrieval Benchmark (PLRB)

PLRB is the retrieval benchmark constructed for this study. The corpus consists of 2,400 source documents drawn from the Philippine OmniCorpus, covering Republic Acts, Supreme Court Decisions, and Administrative Regulations. These documents are segmented into 28,970 overlapping passages of 300 tokens each with a stride of 50 tokens, following BEIR-style passage construction (Thakur et al., 2021) to ensure passage-level coverage of long legal documents.

The query set contains 180 researcher-designed queries: 60 for each of English, Tagalog, and Taglish. Within each language stratum, queries are divided equally across three semantic types—Definitional, Procedural, and Case-Law—informed by historical Philippine Bar Examination question structures.

Train/validation/test split and leakage avoidance: the 28,970-passages index is shared across all phases and is never partitioned, since it represents the fixed retrieval corpus rather than a labeled training set. The 180 PLRB queries were split at the query level into a training split used for adapter triplet sampling and an evaluation split used for retrieval evaluation; no query appears in both splits. Because triplets are constructed from (query, passage) relevance pairs rather than from passages alone, splitting at the query level ensures that no evaluation query's relevance judgments were used to construct training triplets, preventing query-level leakage. The passage index itself is shared between the training and evaluation splits by design, since the retrieval task requires searching the same fixed corpus regardless of which queries are used for adapter training; this is consistent with standard dense-retrieval evaluation protocol, where the corpus is fixed and only the query-side split varies.

Relevance labels were produced through a three-stage pipeline: (1) per-query pooling of the top-10 passages retrieved by each of the three baseline encoders; (2) binary relevance labeling of each query-passage pair by Gemini 2.5 Flash via a structured prompt enforcing legal-relevance criteria; and (3) full research-team review and correction of all LLM-assigned labels prior to finalization. This hybrid pipeline made structured evaluation feasible within undergraduate thesis constraints while subjecting every label to human oversight. The resulting labels are binary (relevant / not relevant) and have not been independently adjudicated by licensed legal professionals—a limitation discussed in Section 7.

3.5 Evaluation Protocol

All metrics reported in Section 5 are computed over the subset of 29 queries (of the 180 total PLRB queries) for which at least one relevant passage exists in the pooled annotation set; queries with no labeled-relevant passages are excluded because Precision@K, Recall@10, and MRR are undefined or trivially zero for them. For each query q_i , let $R_K(q_i)$ denote the set of the top-K retrieved passage IDs and let $G(q_i)$ denote the set of ground-truth relevant passage IDs for q_i .

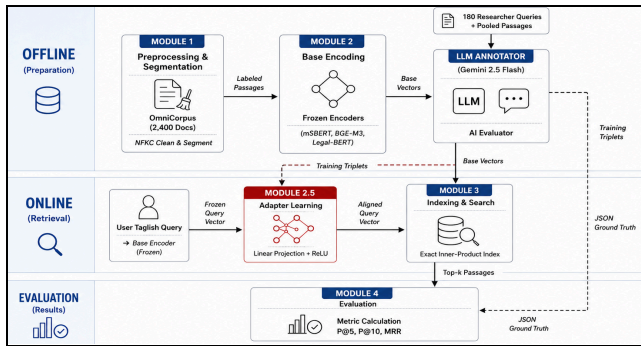
- Precision@K = $|R_K(q_i) \cap G(q_i)| / K$, averaged over all evaluated queries, for $K \in \{5, 10\}$.
- Recall@10 = $|R_{10}(q_i) \cap G(q_i)| / |G(q_i)|$, averaged over all evaluated queries.
- MRR = the mean, over all evaluated queries, of $1 / \text{rank}(q_i)$, where $\text{rank}(q_i)$ is the rank position (1-indexed) of the first retrieved passage that belongs to $G(q_i)$; if no relevant passage is retrieved within the top-10, the reciprocal rank for that query is 0.

All results are reported separately for the baseline (zero-shot) setting and the aligned (adapter) setting, and are further stratified by query language (English, Tagalog, Code-Switched) and by semantic type (Definitional, Procedural, Case-Law).

3.6 System Architecture

The pipeline operates across two phases:

- **Offline Phase —**
 - **Module 1 (Preprocessing and Segmentation):** Unicode NFKC normalization; passage-level language labeling using langdetect with a dual-threshold heuristic for code-switched passages (Fleiss' $\kappa = 0.87$ against manual labels).
 - **Module 2 (Base Encoding):** all 28,970 passages are embedded by each backbone encoder, L2-normalized, and stored as NumPy arrays. Module 2.5 (Adapter Training): the alignment adapter is trained for 50 epochs using Adam on PLRB training triplets.
- **Online Phase —**
 - **Module 3 (Indexing and Search):** FAISS IndexFlatIP structures are built over normalized passage embeddings; at inference, the query is encoded by the frozen base encoder, transformed by the adapter (in the aligned setting), and used for exact inner-product search against the unmodified passage index.
 - **Module 4 (Evaluation):** retrieved passages are compared against PLRB relevance judgments; P@5, P@10, Recall@10, and MRR are reported by language and semantic type.



4. IMPLEMENTATION

4.1 Tools, Technologies, and Environment

The system was implemented in Python 3.12 on a single mid-range GPU workstation. Key library versions are as follows:

- **Sentence-Transformers 2.7.0** — batched vectorization for all three encoder backbones
- **FAISS-GPU 1.7.4** — exact inner-product indexing (IndexFlatIP) over L2-normalized embeddings
- **PyTorch 2.2.1** — adapter training and triplet margin loss optimization
- **Pandas 2.2.1 / PyArrow 15.0.0** — Philippine OmniCorpus .parquet document ingestion and preprocessing
- **Langdetect 1.0.9 with a custom dual-threshold heuristic** — passage-level language identification (English / Tagalog / Code-Switched)
- **Google Vertex AI / Gemini 2.5 Flash** — LLM-assisted relevance annotation via structured prompt
- **MLflow 2.11.1** — experiment tracking for runtimes, memory usage, and evaluation metrics

4.1 Preprocessing Pipeline

Raw, parquet documents from the Philippine OmniCorpus were ingested using PyArrow, deduplicated, null-dropped, and filtered to retain only documents with at least 50 whitespace-delimited tokens. Remaining text was Unicode NFKC-normalized to standardize character representations across documents originating from different digitization sources. Passages were then segmented using a sliding window of 300 tokens with a 50-token stride, yielding 28,970 passages from 2,400 source documents; the 50-token overlap between adjacent passages is intended to reduce the chance that a legally significant clause is split entirely across a passage boundary, though Section 7.3 notes that this risk is not fully eliminated.

Each passage was then assigned a language label—English, Tagalog, or Code-Switched—using langdetect with a custom dual-threshold heuristic: a passage is labeled Code-Switched if langdetect's top-two language probability estimates (for English and Tagalog) both exceed a minimum confidence threshold, rather

than one language dominating; otherwise the passage is labeled with whichever of English or Tagalog receives the higher probability. This heuristic was validated against a manually labeled spot-check sample, achieving Fleiss' $\kappa = 0.87$ agreement with human annotators, indicating substantial but not perfect agreement.

4.3 Development Process

Stage 1 (Weeks 1–2) — Corpus Ingestion and Preprocessing: Raw, parquet documents from the Philippine OmniCorpus were ingested, deduplicated, null-dropped, and filtered as described in Section 4.2. Text was Unicode NFKC-normalized and segmented into 28,970 passages from 2,400 source documents using the 300-token/50-token-stride window. The language-labeling heuristic was applied to every passage, achieving Fleiss' $\kappa = 0.87$ against manual spot-checks.

Stage 2 (Weeks 3–4) — Query Design and Annotation: The 180 PLRB queries were authored by the research team, stratified by language (60 each: English, Tagalog, Taglish) and by semantic type (Definitional, Procedural, Case-Law). Queries were submitted to the LLM-assisted annotation pipeline described in Section 3.4; all Gemini-generated labels were reviewed and corrected by the full research team before finalization.

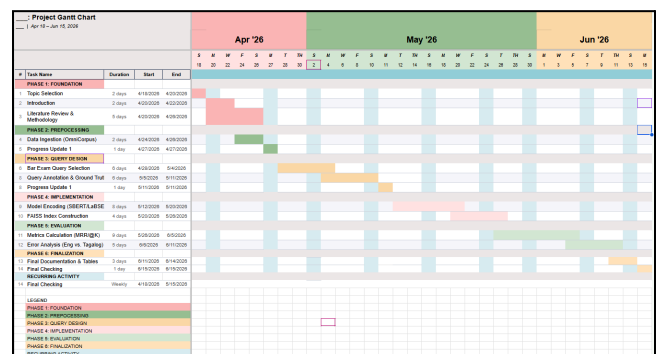
Stage 3 (Weeks 5–6) — Encoder Embedding and Adapter Training: Backbone embeddings for all 28,970 passages were generated and L2-normalized for each of the three encoders. FAISS IndexFlatIP indices were built per encoder and serialized to disk. The alignment adapter (Section 3.3) was then trained for 50 epochs per backbone on the PLRB training-split triplets, with checkpoints saved every 10 epochs for early-stopping analysis.

Stage 4 (Weeks 7–8) — Retrieval Evaluation: All 29 annotated PLRB evaluation queries were encoded and evaluated against relevance judgments under both the zero-shot baseline and adapter-aligned settings, following the protocol in Section 3.5. Results were stratified by language and semantic type, producing the summary and detailed CSV result files referenced throughout Section 5.

4.4 Project Timeline

The development and evaluation of the retrieval system were executed over a structured eight-week lifecycle. A detailed breakdown of task dependencies and milestones is provided in the accompanying Gantt chart.

Sheet Link: [X AI Project_Gantt.xlsx](#)



3. RESULTS AND DISCUSSION

5.1 Results

This section presents the empirical evaluation of BGE-M3, mSBERT, and Legal-BERT under both the zero-shot baseline setting and the adapter-aligned setting on the 29-query PLRB evaluation subset. Performance is measured using MRR, Precision@5 (P@5), Precision@10 (P@10), and Recall@10. All values derive directly from the attached CSV result files.

5.1.1 Overall Retrieval Performance

Table 1 summarizes overall retrieval metrics across all 29 evaluated queries.

Table 1: Overall Retrieval Performance (Baseline vs. Aligned, N = 29)

Model	Setting	MRR	P@5	P@10	Recall@10
BGE-M3	Zero-Shot Baseline	0.8563	0.5310	0.2655	0.7273
	Adapter-Aligned	0.5607	0.2345	0.1483	0.4051
mSBERT	Zero-Shot Baseline	0.4379	0.2552	0.1414	0.2841
	Adapter-Aligned	0.3852	0.1655	0.1069	0.2513
Legal-BERT	Zero-Shot Baseline	0.1506	0.0966	0.0517	0.0878
	Adapter-Aligned	0.0201	0.0138	0.0069	0.0126

BGE-M3 emerges as the dominant architecture in the zero-shot baseline configuration, achieving an MRR of 0.8563, P@5 of 0.5310, P@10 of 0.2655, and Recall@10 of 0.7273. mSBERT shows moderate baseline competence with an MRR of 0.4379. Legal-BERT performs substantially weaker overall with an MRR of 0.1506, reflecting its monolingual training on formal English legal text without any Tagalog or Taglish coverage.

Critically, the adapter does not produce global improvement for any encoder. BGE-M3’s overall MRR falls from 0.8563 to 0.5607 (−34.5%), mSBERT’s from 0.4379 to 0.3852 (−12.0%), and Legal-BERT’s from 0.1506 to 0.0201 (−86.7%). These results are consistent across P@5, P@10, and Recall@10. This aggregate performance drop is the primary finding of this study and indicates overfitting of the adapter to the small training split.

5.1.2 Performance by Query Language

Table 2 presents metric breakdowns by query language. Note that the code-switched stratum contains only one query (Q_013) in the 29-query evaluated subset; statistics for that stratum should be interpreted with caution.

Table 2: Performance Breakdown by Query Language (Baseline vs. Aligned)

(a) Zero-Shot Baseline

Model	Language	MRR	P@5	P@10	Recall@10	N
BGE-M3	Code-Switched	1.0000	0.4000	0.2000	1.0000	1
	English	0.9375	0.6500	0.3250	0.5908	8
	Tagalog	0.8167	0.4900	0.2450	0.7683	20
mSBERT	Code-Switched	0.0000	0.0000	0.0000	0.0000	1
	English	0.6917	0.4750	0.2750	0.5110	8
	Tagalog	0.3583	0.1800	0.0950	0.2075	20
Legal-BERT	Code-Switched	0.0000	0.0000	0.0000	0.0000	1
	English	0.3750	0.2000	0.1125	0.1536	8
	Tagalog	0.0683	0.0600	0.0300	0.0658	20

(b) Adapter-Aligned

Model	Language	MRR	P@5	P@10	Recall@10	N
BGE-M3	Code-Switched	1.0000	0.2000	0.1000	0.5000	1
	English	0.5667	0.2750	0.1875	0.3268	8
	Tagalog	0.5364	0.2200	0.1350	0.4317	20
mSBERT	Code-Switched	0.2500	0.2000	0.1000	0.5000	1
	English	0.7125	0.3500	0.2125	0.3652	8
	Tagalog	0.2611	0.0900	0.0650	0.1933	20
Legal-BERT	Code-Switched	0.0000	0.0000	0.0000	0.0000	1
	English	0.0417	0.0250	0.0125	0.0208	8
	Tagalog	0.0125	0.0100	0.0050	0.0100	20

For code-switched (Taglish) queries, BGE-M3 achieves a perfect MRR of 1.0000 in the zero-shot setting, which is preserved at 1.0000 post-alignment, indicating that the adapter does not disrupt BGE-M3’s strong zero-shot Taglish retrieval on this single example. The most notable positive signal from the adapter is mSBERT’s Taglish performance: its MRR rises from 0.0000 to 0.2500 and P@5 from 0.0000 to 0.2000. This demonstrates that the alignment adapter can recover some Taglish retrieval capability for a weaker multilingual encoder. Legal-BERT scores 0.0000 on Taglish queries in both settings, confirming that monolingual legal-domain pretraining provides no generalization to code-switched text.

For English queries, BGE-M3 leads the zero-shot setting with $MRR = 0.9375$. After alignment, mSBERT is the only encoder to improve on English queries, from 0.6917 to 0.7125, while BGE-M3 degrades sharply from 0.9375 to 0.5667. For Tagalog queries, BGE-M3 leads at baseline ($MRR = 0.8167$), and the adapter degrades performance across all encoders.

5.1.3 Performance by Semantic Query Type

Table 3 stratifies results by query semantic type. All three types—Case-Law, Definitional, and Procedural—draw equally from the 29-query evaluated subset.

Table 3: Performance Breakdown by Semantic Query Type

Model	Semantic Type	Baseline MRR	Baseline P@5	Aligned MRR	Aligned P@5
BGE-M3	Case-Law	0.8750	0.5125	0.5038	0.2125
	Definitional	0.8542	0.6000	0.6500	0.3000
	Procedural	0.8000	0.4800	0.6000	0.2000
mSBERT	Case-Law	0.3646	0.2250	0.3154	0.1750
	Definitional	0.5625	0.3000	0.5781	0.2000
	Procedural	0.4733	0.2800	0.3000	0.0800
Legal-BERT	Case-Law	0.0625	0.0250	0.0000	0.0000
	Definitional	0.1667	0.1250	0.0417	0.0250
	Procedural	0.4067	0.2800	0.0500	0.0400

BGE-M3 achieves the highest baseline MRR across all types: 0.8750 (Case-Law), 0.8542 (Definitional), 0.8000 (Procedural). Adapter alignment reduces BGE-M3’s performance across all three types, with the largest absolute drop in Case-Law (0.8750 $\rightarrow$ 0.5038). mSBERT shows its strongest adaptation gain on Definitional queries (MRR: 0.5625 $\rightarrow$ 0.5781), the sole improvement for mSBERT other than Taglish.

For Procedural queries, Legal-BERT achieves its strongest zero-shot baseline ($MRR = 0.4067$), likely because procedural legal text is more formulaic and English legal pretraining provides useful overlap. However, adapter alignment causes a catastrophic drop to $MRR = 0.0500$, further confirming that the adapter degrades Legal-BERT across all conditions except where zero-shot performance was already near zero.

5.2 Discussion

The results yield two main findings. First, modern multilingual encoders—particularly BGE-M3—already provide strong zero-shot retrieval for Philippine legal queries, even for Tagalog and code-switched inputs. BGE-M3’s zero-shot MRR of 0.8563 overall and 1.0000 on the Taglish query indicate that its broad cross-lingual pretraining captures sufficient semantic overlap between informal Filipino-English queries and formal English statutes to rank relevant passages highly. This finding supports the

viability of zero-shot multilingual retrieval as a practical baseline for Philippine legal information access.

Second, the neural alignment adapter does not generalize beyond the training distribution under current data conditions. The aggregate performance degradation across all three encoders and all metric dimensions points to overfitting as the primary explanation. With only 180 total queries (of which a subset formed the adapter training split), the adapter learns query-specific patterns that do not transfer to held-out queries. The adapter also used uniform random negatives rather than hard negatives, which may have produced an insufficiently discriminative training signal for already-strong encoders such as BGE-M3.

The one exception—mSBERT’s Taglish MRR improving from 0.0000 to 0.2500—demonstrates that the adapter can recover meaningful Taglish retrieval capability for encoders that lack it in zero-shot mode. This suggests that lightweight alignment is most valuable for weaker multilingual baselines or in settings where zero-shot performance is near chance on code-switched queries.

Qualitatively, failure cases for mSBERT on Tagalog queries (e.g., Q_033, Q_139, Q_153, all $MRR = 0.0000$ at baseline) reveal a consistent pattern: the top-10 retrieved passages cluster around a small set of frequently retrieved legal passages (e.g., PASSAGE_27670, PASSAGE_10232, PASSAGE_10150), suggesting that mSBERT’s embedding space collapses Tagalog queries toward a narrow region regardless of their specific legal content. This vocabulary-distribution issue is not addressed by the current adapter, which inherits the base encoder’s embedding geometry.

6. CONCLUSION AND FUTURE WORK

6.1 Conclusion

This study investigated the embedding mismatch between informal Taglish legal queries and formal English Philippine legal passages through two mechanisms: the design of a parameter-efficient neural alignment adapter, and the introduction of the first Philippine Legal Retrieval Benchmark (PLRB). Three encoder architectures—BGE-M3, mSBERT, and Legal-BERT—were evaluated under both zero-shot and adapter-aligned settings on 29 annotated PLRB queries.

The principal practical takeaway is that zero-shot multilingual retrieval—specifically BGE-M3—already provides a strong and reliable retrieval framework for Philippine legal queries, achieving an overall MRR of 0.8563 without any additional training. The neural alignment adapter, by contrast, reduced overall performance for all three encoders due to overfitting on the small training split. The sole positive signal from the adapter was the recovery of mSBERT’s Taglish retrieval from $MRR = 0.0000$ to 0.2500, demonstrating that lightweight alignment can help weaker baselines under resource constraints.

Legal-BERT’s consistent failure on Taglish and Tagalog queries—0.0000 MRR in both settings—confirms that legal-domain pretraining alone is insufficient when the encoder lacks multilingual or code-switching capability. Institutions seeking to deploy Philippine legal retrieval systems today should prioritize broad multilingual encoders such as BGE-M3 in zero-shot mode over lighter legal-domain models.

The PLRB benchmark contributes a structured evaluation resource—2,400 documents, 28,970 passages, 180 queries—for

future Philippine legal NLP and retrieval research, even as its current scale limits adapter generalization.

6.2 Future Work

The observed weaknesses point directly to the following priorities for future research:

- Expand PLRB query volume substantially (e.g., to 1,000+ queries per language) to provide a training split large enough to prevent adapter overfitting and to yield stable evaluation estimates for the code-switched stratum.
- Incorporate expert legal annotation. Licensed Philippine legal practitioners should review and adjudicate relevance labels to replace or supplement the current LLM-assisted labels, improving benchmark validity and reducing annotator-bias risk.
- Adopt hard negative mining during adapter training to provide more discriminative triplets and reduce the risk that the adapter learns superficial query-specific patterns.
- Explore stronger parameter-efficient adaptation methods, including LoRA, prefix tuning, and multilingual instruction-tuned embedding models, to determine whether superior alignment can be achieved without full fine-tuning.
- Investigate hybrid retrieval architectures combining dense semantic retrieval with sparse BM25 to recover performance on highly specific legal terminology that dense encoders may miss.
- Extend PLRB to Philippine regional languages (Cebuano, Ilocano, Hiligaynon, Bikol) to better reflect the multilingual reality of Philippine legal information access.
- Conduct reranking experiments using cross-encoders or rerankers such as bge-reranker-m3 as a post-retrieval stage, to separate first-stage retrieval quality from final ranking quality.

7. LIMITATIONS OF THE STUDY

This study was conducted under strict time constraints; the following limitations should be considered when interpreting results.

7.1 Query and Label Authorship Bias

All 180 PLRB queries were written by the research team, who also designed the retrieval system and selected the training objective. This introduces potential confirmatory bias: the team's familiarity with the legal passages may have influenced query phrasing in ways that inadvertently favor the retrieval system's

known strengths. Queries were not independently sourced from actual legal searchers (e.g., law students, paralegals), which limits ecological validity.

Relevance labels were generated by Gemini 2.5 Flash and subsequently reviewed by the research team. This review was not conducted by licensed Philippine legal professionals and therefore does not constitute formal expert adjudication. LLMs can misinterpret subtle legal distinctions, apply inconsistent relevance standards across query types, or conflate factually adjacent passages. The binary label format further limits nuanced relevance grading. Benchmark claims should be treated as preliminary until independently validated by legal experts.

7.2 Scale and Scope

The evaluation subset of 29 queries is small, and performance estimates—particularly for the code-switched stratum, which contains only one query—are statistically unreliable. MRR computed over a single query is equivalent to the reciprocal of that query's rank, not a distributional aggregate. Results for the Taglish stratum should be treated as illustrative case evidence rather than generalizable statistics.

The 180-query total also constrained adapter training: the effective training split was insufficient to prevent overfitting for all three encoders, as evidenced by the universal performance decline post-alignment. Scaling the benchmark by an order of magnitude is a prerequisite for meaningful adapter training.

7.3 Domain-Shift and Pipeline Risks

The PLRB corpus draws exclusively from three document types (Republic Acts, Supreme Court Decisions, Administrative Regulations) from the Philippine OmniCorpus. Legal passages from other source types (local ordinances, agency memoranda, bar association advisories) may exhibit different language patterns and retrieval difficulty profiles. The passage segmentation heuristic (300-token windows with 50-token stride) may split legally significant reasoning across passage boundaries, artificially lowering retrieval recall.

The language detection heuristic (langdetect with custom dual-threshold) achieves Fleiss' $\kappa = 0.87$ but is not infallible; misclassified passages could introduce noise into language-stratified evaluations. Domain-shift risk also applies to the annotation pipeline: Gemini 2.5 Flash was prompted with general legal relevance criteria rather than Philippine-jurisdiction-specific standards, which may have introduced systematic label errors for jurisdiction-specific provisions.

REFERENCES

- Ahmadi, N. B., Strich, J., Semmann, M., & Biemann, C. (2026). LEMUR: A corpus for robust fine-tuning of multilingual law embedding models for retrieval. arXiv. <https://doi.org/10.48550/arXiv.2602.09570>
- Aquino, A., & de Leon, M. (2020). Developing a named entity recognition corpus for Filipino. In Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020) (pp. 3637–3645). European Language Resources Association.

- Barrios, J., Cruz, J. C. B., & Cheng, C. (2023). Developing a named entity recognition dataset for Tagalog. arXiv. <https://doi.org/10.48550/arXiv.2311.07161>
- Chalkidis, I., Fergadiotis, M., Malakasiotis, P., & Androutsopoulos, I. (2022). Large-scale multi-label text classification on EU legislation. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL 2022) (pp. 6661–6683). Association for Computational Linguistics. <https://doi.org/10.18653/v1/P19-1669>
- Cruz, J. C. B., & Cheng, C. (2019). Evaluating language model finetuning techniques for low-resource languages. arXiv. <https://doi.org/10.48550/arXiv.1907.00409>
- Cruz, J. C. B., & Cheng, C. (2021). Improving large-scale language models and resources for Filipino. arXiv. <https://doi.org/10.48550/arXiv.2111.06053>
- Department of Science and Technology – Advanced Science and Technology Institute. (2021). iTANONG: A Taglish natural language interface for database querying (Technical Report). DOST-ASTI.
- Kim, M.-Y., Rabelo, J., Babiker, H. K. B., Rahman, M. A., & Goebel, R. (2024). Legal information retrieval and entailment using transformer-based approaches. *The Review of Socionetwork Strategies*, 18(1), 101–121. <https://doi.org/10.1007/s12626-023-00153-7>
- Martin, I. P. (2012). Expanding the role of Philippine languages in the legal system. *Perspectives in the Arts and Humanities Asia*, 2(1), 1–20.
- Schroff, F., Kalenichenko, D., & Philbin, J. (2015). FaceNet: A unified embedding for face recognition and clustering. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2015) (pp. 815–823). IEEE. <https://doi.org/10.1109/CVPR.2015.7298682>
- Supreme Court of the Philippines. (2022a). Chief Justice Gesmundo: Judiciary e-library to use AI technology to improve legal research [Press release]. Supreme Court of the Philippines Public Information Office.
- Supreme Court of the Philippines. (2022b). Strategic plan for judicial innovations 2022–2027. Supreme Court of the Philippines.
- Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., & Gurevych, I. (2021). BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS 2021) Datasets and Benchmarks Track. Curran Associates.
- Voorhees, E. M. (2001). The philosophy of information retrieval evaluation. In Proceedings of the 2nd Workshop of the Cross-Language Evaluation Forum (CLEF 2001) (LNCS 2406, pp. 355–370). Springer. https://doi.org/10.1007/3-540-45691-0_34
- Wang, L., Yang, N., Huang, X., Jiao, B., Yang, L., Jiang, D., Majumder, R., & Wei, F. (2022). Text embeddings by weakly-supervised contrastive pre-training. arXiv. <https://doi.org/10.48550/arXiv.2212.03533>